

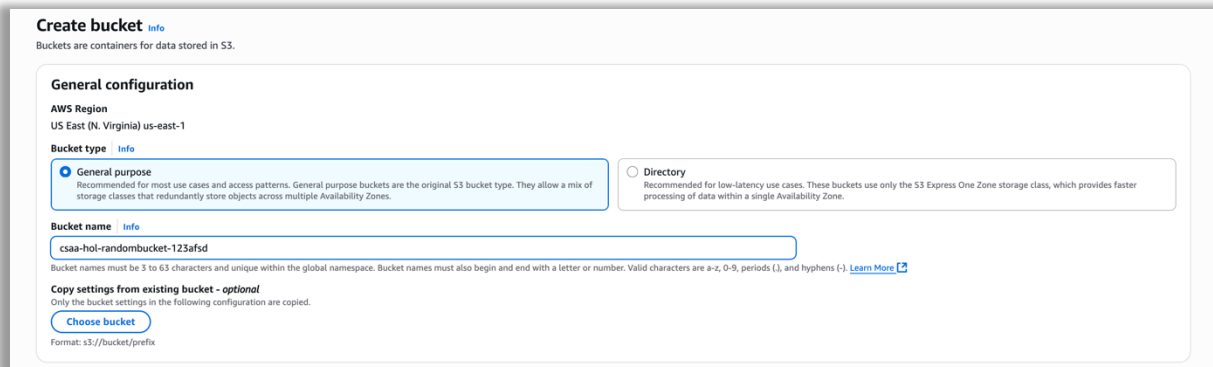
Sending VPC Flow Logs to S3 and Parsing the Logs via Amazon Athena

Chris Samugana

Introduction

In this project, I implemented a solution using Amazon VPC Flow Logs, Amazon S3, and Amazon Athena to build a serverless analytics platform for querying network traffic logs with SQL-like syntax. The solution helps improve visibility into network behavior, supports security auditing, and facilitates operational troubleshooting.

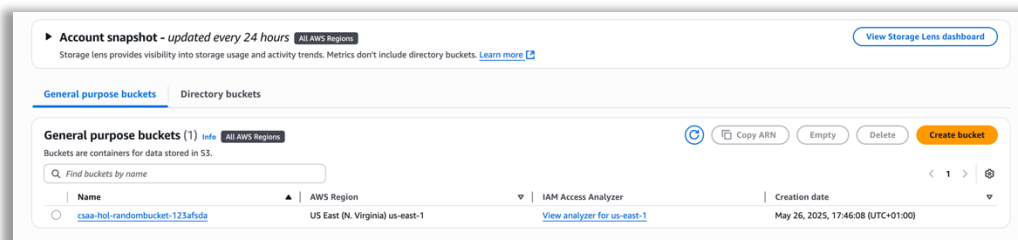
Solution Overview



The screenshot shows the 'Create bucket' page in the AWS S3 console. It includes sections for 'General configuration', 'Bucket type' (General purpose selected), 'Bucket name' (csaa-hol-randombucket-123afsd), and 'Copy settings from existing bucket - optional'. A 'Choose bucket' button is at the bottom.

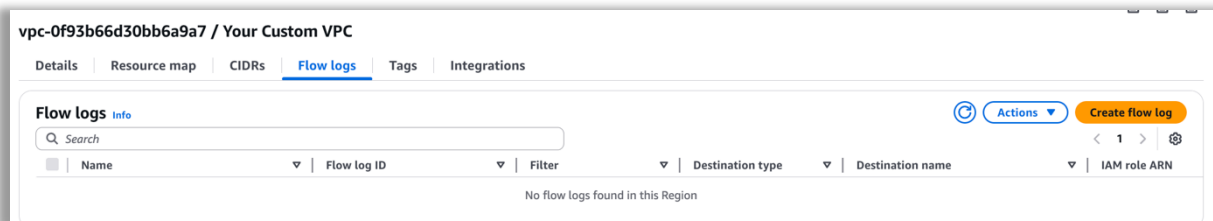
1. Creating the Amazon S3 Bucket

- Accessed the Amazon S3 console and navigated to the Buckets section.
- Created a new bucket in the us-east-1 Region with the required prefix csaa-hol-.
- Kept all default settings and successfully created the bucket for log storage.

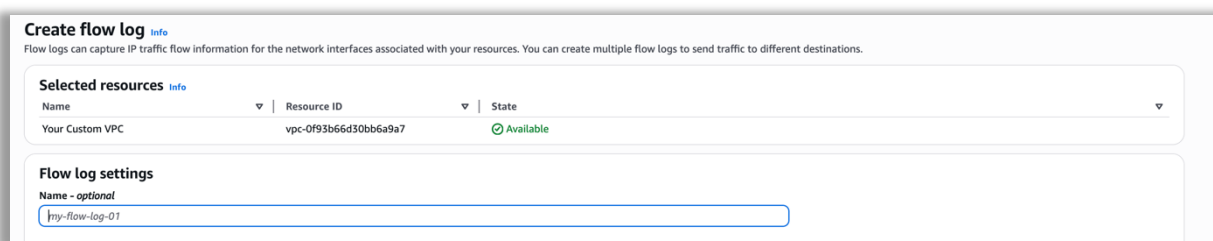


The screenshot shows the 'General purpose buckets' section of the AWS S3 console. It displays a table with one bucket: 'csaa-hol-randombucket-123afsd' in the 'US East (N. Virginia) us-east-1' region, created on 'May 26, 2025, 17:46:08 (UTC+01:00)'. A 'View analyzer for us-east-1' link is provided.

2. Creating the VPC Flow Log and Generating Traffic



The screenshot shows the 'Flow logs' section of the AWS VPC console for a VPC named 'vpc-0f93b66d30bb6a9a7'. It displays a table with no flow logs found in this region. A 'Create flow log' button is visible.



The screenshot shows the 'Create flow log' page in the AWS VPC console. It includes sections for 'Selected resources' (vpc-0f93b66d30bb6a9a7) and 'Flow log settings' (Name: my-flow-log-01).

- Opened the Amazon VPC console and selected Your VPCs.
- Located my custom VPC and navigated to its Flow Logs section.
- Created a new flow log with the following parameters:
 - Name: vpc-to-s3
 - Filter: All
 - Aggregation interval: 1 minute
 - Destination: Amazon S3
 - S3 ARN: Copied from my newly created bucket
 - Log record format: AWS default (version 2)
 - Log file format: Parquet
 - Hive-compatible prefix: Enabled
 - Partition logs by time: Every 1 hour
- Deployed the flow log successfully and confirmed it was attached to my VPC.

To generate traffic:

- Navigated to Amazon EC2 > Load Balancers and located the DNS name of OurApplicationLoadBalancer.
- Accessed the DNS address using HTTP and refreshed multiple times to generate sample flow logs.
- Waited several minutes for the logs to propagate into the S3 bucket.
- Verified the S3 bucket began to populate with objects under the AWSLogs/ prefix.

Create flow log Info

Flow logs can capture IP traffic flow information for the network interfaces associated with your resources. You can create multiple flow logs to send traffic to different destinations.

Selected resources Info		
Name	Resource ID	State
Your Custom VPC	vpc-0f93b6d30bb6a9a7	Available

Flow log settings

Name - optional

✓ Bucket ARN copied

↺
📄 Copy ARN
Empty
Delete
Create bucket

S3 bucket ARN

The ARN of the Amazon S3 bucket to which the flow log is published. You can specify a specific folder in the bucket using the bucket_ARN/folder_name/ format. [Create S3 bucket](#)

ⓘ Please note, a resource-based policy will be created for you and attached to the target bucket.

Log file format Info

The format for the log files. Each log file is compressed using Gzip compression.

☐ Text (default)
 ☒ Parquet

Hive-compatible S3 prefix [Info](#)

Enable to use Hive-compatible S3 prefixes to simplify the loading of new data into your Hive-compatible tools.

☒ Enable

Partition logs by time [Info](#)

Partition your logs per hour to reduce your query costs and get faster response if you have a large volume of logs a

☐ Every 24 hours (default)

☒ Every 1 hour (60 minutes)

Successfully created flow log for vpc-0f93b66d30bb6a9a7.

Explore integration options

Your VPCs (1/1) [Info](#)

Last updated 4 minutes ago [Actions](#) [Create VPC](#)

Find VPCs by attribute or tag

☒	Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set
☒	Your Custom VPC	vpc-0f93b66d30bb6a9a7	Available	Off	10.0.0.0/16	-	dopt-04a5984821

vpc-0f93b66d30bb6a9a7 / Your Custom VPC

Details | Resource map | CIDRs | [Flow logs](#) | Tags | Integrations

Flow logs (1) [Info](#)

[Actions](#) [Create flow log](#)

Search

☐	Name	Flow log ID	Filter	Destination type	Destination name	IAM role ARN
☐	vpc-to-s3	fl-0d9208b37adf35b	ALL	s3	csaa-hol-randombucket-123afsd	-

Load balancers (1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in in

Filter load balancers

☐

Name

☒

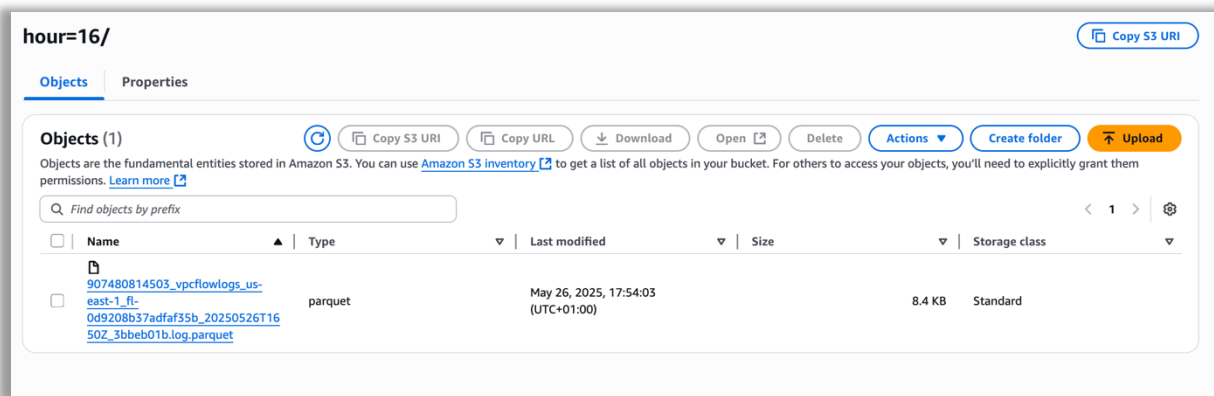
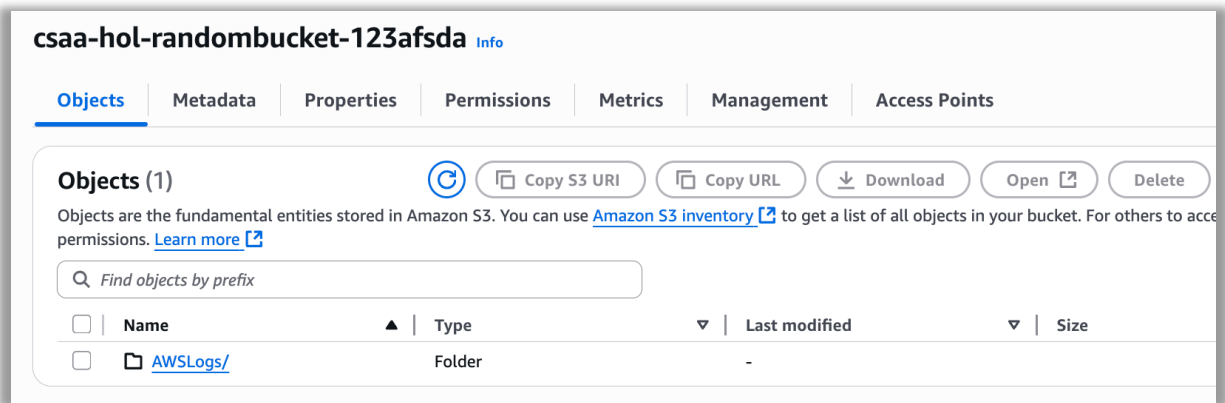
OurApplicationLoadBal...

☐

OurApplicationLoadBalance...

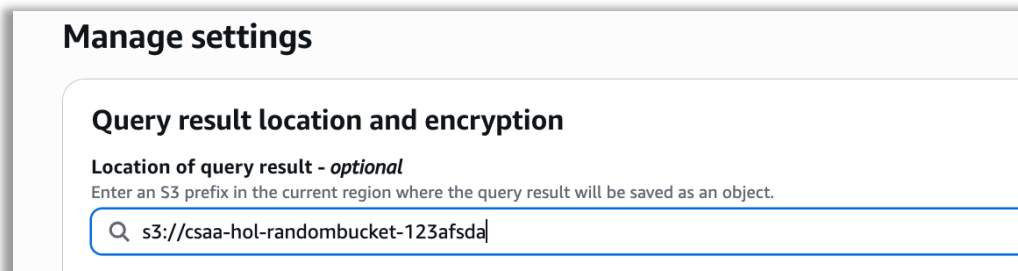
Active

DNS name copied



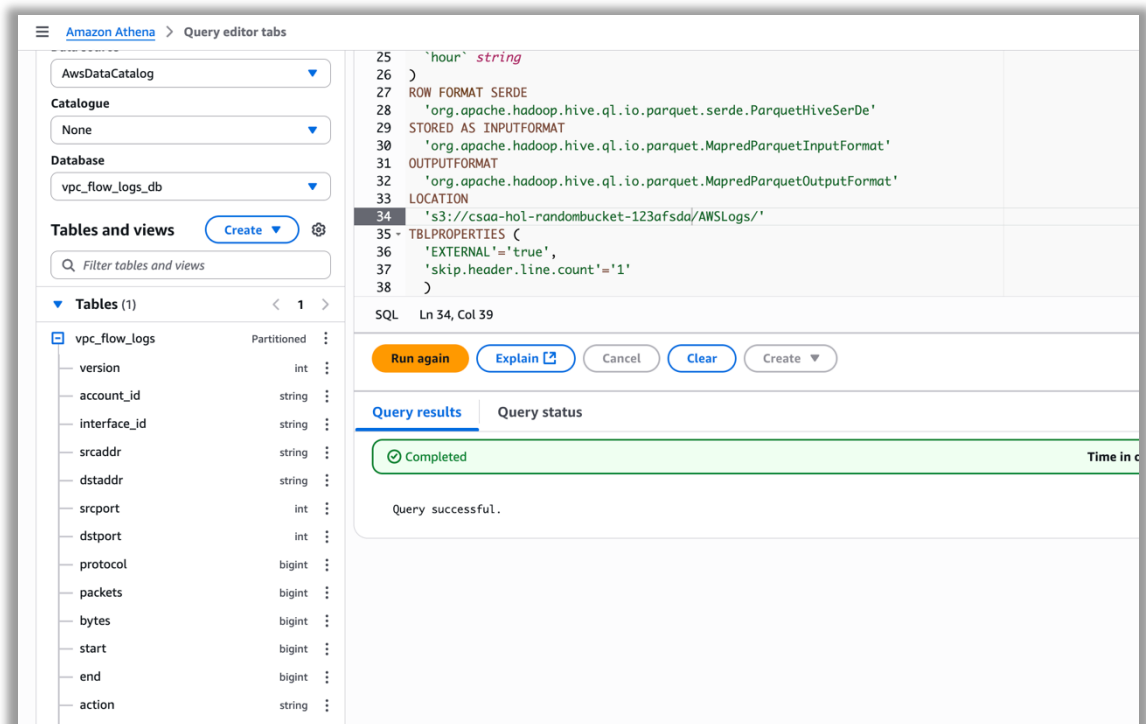
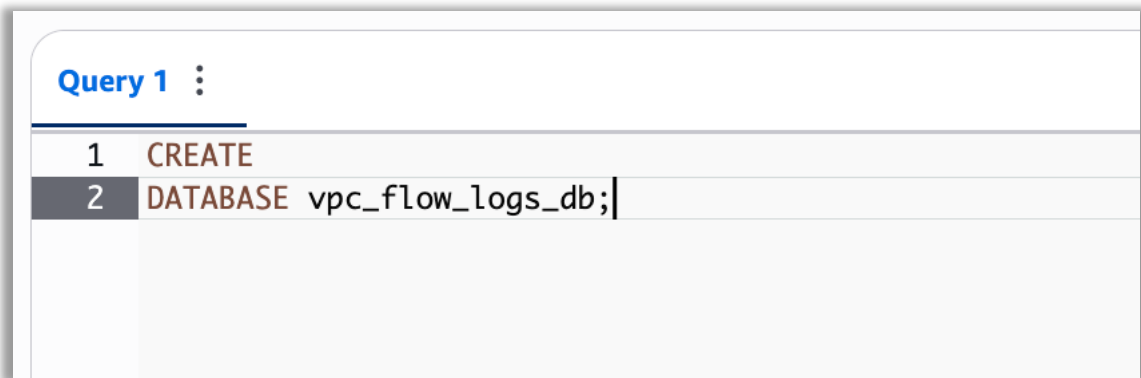
3. Creating the Amazon Athena Table

- Opened Amazon Athena, navigated to Query Editor.
- Configured the Query result location by pointing it to my S3 bucket and appending /queries/.



CREATE DATABASE vpc_flow_logs_db;

- Selected the newly created vpc_flow_logs_db database from the left-hand panel.
- Created a new table with the following modified SQL query (updating BUCKET_NAME with my actual S3 bucket name):
- Successfully ran the table creation query and then executed the following to load the partitions and make the table queryable:



4. Querying the Flow Logs Data

- With the table ready, executed a sample query to retrieve flow log records:
- Successfully retrieved and reviewed records from the VPC Flow Logs, confirming the data pipeline was working from log generation in VPC → S3 → Athena.

Query 1Query 2Query 3

1 MSCK

2 REPAIR TABLE vpc_flow_logs;

SQLLn 2, Col 28

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery status

Completed

Time in queue: 199 msRun time: 4.377 secData scanned: -

Partitions not in metastore: vpc_flow_logs:aws-account-id=907480814503/aws-service=vpcflowlogs/aws-region=us-east-1/year=2025/month=05/day=26/hour=16
Repair: Added partition to metastore vpc_flow_logs:aws-account-id=907480814503/aws-service=vpcflowlogs/aws-region=us-east-1/year=2025/month=05/day=26/hour=16

Query 1Query 2Query 3Query 4

1 SELECT *

2 FROM vpc_flow_logs LIMIT 10;

SQLLn 2, Col 29

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery status

Completed

Time in queue: 104 msRun time: 1.28 secData scanned: 10.02 KB

Results (10)

CopyDownload results CSV

Search rows

#	version	account_id	interface_id	srcaddr	dstaddr	srcport	dstport	protocol	packets	bytes	s
1	2	907480814503	eni-00332708f14d9c2f0	167.94.146.47	10.0.2.184	7227	63630	6	1	60	1
2	2	907480814503	eni-00332708f14d9c2f0	10.0.10.169	10.0.2.184	80	49540	6	5	696	1
3	2	907480814503	eni-00332708f14d9c2f0	10.0.2.184	10.0.10.169	49540	80	6	6	446	1
4	2	907480814503	eni-00332708f14d9c2f0	10.0.11.216	10.0.2.184	80	38248	6	5	696	1
5	2	907480814503	eni-00332708f14d9c2f0	10.0.2.184	10.0.11.216	38248	80	6	5	394	1
6	2	907480814503	eni-00332708f14d9c2f0	10.0.12.142	10.0.2.184	80	16714	6	5	696	1
7	2	907480814503	eni-00332708f14d9c2f0	10.0.2.184	10.0.12.142	16714	80	6	5	394	1
8	2	907480814503	eni-00332708f14d9c2f0	206.168.35.28	10.0.2.184	31846	20201	6	1	60	1
9	2	907480814503	eni-00332708f14d9c2f0	47.82.103.16	10.0.2.184	36399	1027	17	1	116	1
10	2	907480814503	eni-00332708f14d9c2f0	64.62.156.54	10.0.2.184	38622	2121	6	1	40	1

Conclusion

This project demonstrated the end-to-end implementation of a serverless logging analytics platform using AWS-native services. I successfully:

- Configured VPC Flow Logs to deliver data to Amazon S3 in Parquet format.
- Set up Amazon Athena to query flow log data efficiently using Hive-compatible partitioning.
- Validated real-time network traffic capture and visibility through Athena queries.

This architecture enables scalable, low-maintenance log analysis without provisioning dedicated servers, supporting cloud-native observability and operational intelligence.